Sumanth Kothamasu

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PROFESSIONAL SUMMARY:

Senior Java Full Stack Developer with 10+ years of experience designing and building scalable distributed systems and cloud-native microservices across banking, healthcare, e-commerce, and automotive domains. Deep expertise in Java (8/11), Spring Boot, and microservices architecture, with proven ability to develop high-performance RESTful APIs and scalable event-driven systems using Apache Kafka. Extensive hands-on experience with AWS cloud infrastructure, containerization and orchestration using Docker and Kubernetes, and CI/CD automation through Jenkins and GitHub Actions. Specialized in implementing secure authentication and authorization using OAuth2 and JWT and building enterprise-grade systems compliant with PCI-DSS, HIPAA, and SOX standards. Contributed to critical enterprise initiatives including microservices architecture design, cloud-native deployment strategies, and performance optimization through caching (Redis), database tuning (PostgreSQL/MySQL), and architectural improvements, consistently delivering high-availability, low-latency distributed systems. Strong collaborator across engineering, QA, DevOps, and product teams with consistent delivery in Agile environments.

TECHNICAL SKILLS:

- **Programming Languages:** Java (8,11, 17), JavaScript (ES6+)
- **Backend Technologies & Frameworks:** Spring Boot, Spring MVC, Spring Data JPA, Hibernate ORM
- **Microservices & Architecture:** RESTful APIs, Microservices Design Patterns, Event-Driven Architecture, Service Decomposition, API Versioning, Circuit Breaker (Resilience4j), Saga Pattern, Distributed Systems (High Availability, Low Latency, Fault Tolerance)
- **Security & Authentication:** Spring Security, OAuth2, JSON Web Tokens (JWT), RBAC
- **Compliance & Standards:** PCI-DSS, HIPAA, HL7, FHIR, GLBA
- **Frontend Technologies**: React, Angular, Typescript, HTML, CSS
- **State Management & UI Architecture:** Redux, Context API
- **Cloud & Infrastructure:** Amazon Web Services (AWS) – EC2, S3, EKS, IAM, CloudWatch, RDS, ALB, VPC
- **DevOps & CI/CD:** Jenkins, GitHub Actions, CI/CD pipeline, Git, Gitflow
- **Caching & Performance:** Redis, Performance Tuning, Performance Profiling
- **API & Documentation Tools:** Swagger, OpenAPI, Rest Assured
- **Containerization & Orchestration:** Docker, Kubernetes
- **Messaging & Streaming:** RabbitMQ, Apache Kafka
- **Build & Deployment Tools:** Maven, Apache Tomcat
- **Testing & Quality Assurance:** Jest, JUnit, Mockito, Spring Boot Test
- **Logging & Monitoring:** ELK Stack (Elasticsearch, Logstash, Kibana), Splunk, AWS CloudWatch, Prometheus, Grafana
- **Distributed Tracing:** Zipkin, Open Telemetry
- **Databases & Data Management:** MySQL, PostgreSQL, Oracle, SQL, NoSQL, Schema Design, Transaction Management, Query Optimization, Database Indexing, Replication, Connection Pooling.
- **Development Practices & Methodologies:** Agile Scrum, Sprint Planning, Backlog Grooming, Retrospectives
- **Other tools:** JMeter, SonarQube, Postman, JIRA, Confluence, IntelliJ IDEA, Eclipse, VS Code

PROFESSIONAL EXPERIENCE:

Fifth Third Bank, Evansville, IN — **Senior Java Full Stack Developer | Nov 2024 – Present**

Roles & Responsibilities:

- Developed microservices for Fifth Third's online banking platform using Java 17, Spring Boot 3, and React.js with TypeScript, supporting retail and commercial banking operations including account management, payment processing, wire transfers, and bill pay functionality serving 2M+ customers.
- Built RESTful APIs for account-service and payment-processing-service, implementing account creation, balance inquiries, transaction validation, payment confirmation, and fund transfer workflows handling thousands of transactions per second during peak banking hours.
- Implemented payment processing workflows for multiple payment types including ACH transfers, wire payments, bill payments, and internal account transfers, integrating with third-party payment processors (FIS, Fiserv, ACH networks) and processing $500M+ monthly transaction volume.
- Implemented retry logic with exponential backoff, circuit breaker pattern (Resilience4j), and idempotency key validation to prevent duplicate transactions and ensure payment reliability across distributed systems.
- Optimized account dashboard and mobile app performance by analysing PostgreSQL slow query logs, identifying N+1 query patterns, implementing composite indexes on account_id and transaction_date columns, and applying React code splitting with lazy loading and Webpack bundle optimization, reducing page load times from 5 seconds to 2 seconds.
- Implemented Redis caching for frequently accessed account summaries, current balances, and recent transaction history with appropriate TTL strategies, reducing database read queries by approximately 60% and improving API response times from 450ms to 140ms during peak hours.
- Worked within database-per-service architecture, implementing optimistic locking for concurrent account updates and pessimistic locking for critical balance checks to prevent race conditions and ensure data integrity in payment workflows.

- Contributed to event-driven payment workflows using Apache Kafka, implementing saga pattern steps for distributed transactions coordinating account debits, payment processing, fraud checks, and notifications across microservices
- Developed modular React.js frontend components using TypeScript for account management features including account dashboard displaying balances and recent transactions, transaction history with date range filtering, fund transfer forms with real-time validation, and bill payment scheduling interface.
- Implemented Redux for state management in multi-step workflows including wire transfers with beneficiary selection, amount entry, and confirmation screens, and used Context API for maintaining authentication state and account context across components.
- Integrated backend APIs with comprehensive error handling for banking-specific scenarios (insufficient funds, invalid accounts), loading states with skeleton screens, retry logic with exponential backoff for network failures, and optimistic UI updates for instant feedback.
- Implemented Spring Security with OAuth2 and JWT for authentication and authorization, developed secure RESTful APIs with comprehensive input validation, SQL injection prevention, XSS protection, and CSRF tokens, ensuring compliance with PCI-DSS, SOX, and GLBA standards.
- Implemented timeout policies for external service calls with 3-second connection and 10-second read timeouts and ensured secure data handling using TLS 1.3 for data in transit and AES-256 encryption at rest for PCI-DSS compliance.
- Worked with DevOps team on deploying microservices to AWS infrastructure with auto-scaling configurations ranging from 3-5 instances for account-service to 4-6 instances for payment-processing-service, leveraging multi-AZ RDS deployment with read replicas for high-availability account operations.
- Containerized microservices using Docker with multi-stage builds and implemented health check endpoints for Kubernetes liveness and readiness probes, enabling zero-downtime deployments on AWS EKS
- Contributed to CI/CD pipeline enhancements using Jenkins, automating Maven builds, SonarQube code quality scanning to check code coverage, Docker image creation, Trivy vulnerability scanning, and automated deployment to AWS EKS with rollback capabilities.
- Integrated account and payment services with centralized ELK Stack (Elasticsearch, Logstash, Kibana) implementing structured logging with correlation IDs propagated across all microservices, enabling end-to-end distributed request tracing from initial API call through Kafka events to final database update.
- Developed comprehensive test suites using JUnit 5, Mockito, Spring Boot Test, and REST Assured, achieving 85% code coverage for account and payment business logic and ensuring reliability of critical banking workflows.
- Provided technical guidance to 2 junior developers, conducting code reviews focused on Spring Boot microservices patterns for account and payment workflows, secure coding practices (OWASP Top 10), PCI-DSS compliance requirements, and idempotency handling, reducing their pull request review cycles from 3-4 rounds to 1-2 rounds.
- Collaborated with cross-functional team of 8-10 engineers including backend developers, frontend engineers, DevOps team members, security engineers, and QA testers in Agile Scrum environment with 2-week sprints, delivering banking features on schedule.

Environment: Java 17, Spring Boot 3, Spring MVC, Spring Data JPA, Spring Security, Hibernate ORM, Resilience4j, Microservices Architecture, RESTful APIs, React.js, TypeScript, JavaScript (ES6+), Redux, Context API, OAuth2, JWT, PostgreSQL, Apache Kafka, Redis, HikariCP, AWS (EC2, S3, RDS, EKS, CloudWatch, ALB, Auto Scaling, VPC), Docker, Kubernetes, Jenkins, CI/CD, Maven, ELK Stack, SonarQube, Trivy, JUnit 5, Mockito, Spring Boot Test, REST Assured, Git, Agile Scrum, PCI-DSS, SOX, GLBA Compliance.

Centene Corporation, Saint Louis, MO **Senior Java Full Stack Developer | Oct 2022 - Oct 2024**
Roles & Responsibilities:

- Developed full-stack healthcare applications using Java 11, Spring Boot, and React.js for member enrollment, provider search, and benefits management workflows supporting Medicaid and Medicare operations.
- Contributed to microservices architecture development for member portal platform, building independent services for member-service, provider-service, and eligibility-service with separate deployments, enabling parallel development across teams and improving system scalability.
- Developed member-service microservice handling member profile operations including registration, authentication, profile updates, and preference management, implementing service-to-service communication with eligibility-service and claims-service using REST APIs and asynchronous messaging.
- Implemented provider-service microservice for provider directory management, handling provider search, filtering by specialty and location, network status verification, and real-time availability checks, serving 100K+ provider search requests daily.
- Implemented Spring Security with OAuth2 and JWT for authentication and authorization across microservices, enforcing role-based access control to protect sensitive member and claims data and ensure HIPAA compliance.
- Optimized SQL queries and implemented indexing strategies in MySQL for member demographics, provider information, and claims history data to improve query performance for high-traffic member portal microservices.
- Developed responsive React.js frontend applications using JavaScript (ES6+), HTML5, CSS3, Redux, and Context API for member self-service portals, consuming REST APIs from multiple backend microservices to display unified member dashboard.

- Integrated frontend with multiple backend microservices using RESTful APIs, implementing comprehensive error handling for healthcare-specific scenarios (eligibility not found, claims processing delays), loading states with skeleton screens, and retry logic for network failures.
- Implemented Redis caching for frequently accessed member eligibility data and provider directory information to reduce database load and improve API response times from member-service and provider-service during peak traffic periods.
- Developed inter-service communication patterns using Apache Kafka for event-driven workflows, publishing events for member enrollment updates, eligibility changes, and provider updates, enabling loose coupling between microservices and improving system resilience.
- Implemented Resilience4j-based patterns including circuit breakers, retry mechanisms, and fallback strategies in microservices to improve system stability when communicating with downstream services and external healthcare partner systems.
- Containerized microservices using Docker with multi-stage builds for member-service, provider-service, and eligibility-service, enabling consistent deployments across development, testing, and production environments.
- Worked with DevOps team on deploying microservices to AWS infrastructure, leveraging EC2 instances with auto-scaling capabilities and Application Load Balancers to handle variable workloads during open enrollment periods for Medicaid and Medicare operations.
- Contributed to CI/CD pipeline implementation using Jenkins, supporting Maven builds, Docker image creation, automated testing, and deployment of microservices to AWS infrastructure for faster release cycles.
- Implemented centralized logging using ELK Stack and AWS CloudWatch across all microservices with correlation IDs for distributed request tracing, enabling end-to-end monitoring of requests flowing through member-service, eligibility-service, and claims-service.
- Worked on implementing distributed tracing using OpenTelemetry to track request flows across microservices, enabling identification of performance bottlenecks and faster debugging of member enrollment and claims processing workflows.
- Developed comprehensive unit and integration tests using JUnit and Mockito for microservices, improving test coverage for member portal and eligibility verification services and ensuring reliability of healthcare workflows.
- Created API documentation using Swagger/OpenAPI for each microservice, documenting endpoints, authentication requirements, request/response schemas, and service dependencies for internal team collaboration and external partner integrations.
- Collaborated with cross-functional teams including backend developers, QA engineers, DevOps, business analysts, and compliance teams in Agile Scrum environment with 2-week sprints to deliver member-facing microservices aligned with healthcare regulations.

Environment: Java 11, Spring Boot, Microservices, RESTful APIs, Spring Security, Spring Data JPA, React.js, Redux, Context API, JavaScript ES6+, HTML5, CSS3, MySQL, Apache Kafka, AWS (EC2, S3, RDS, EKS, ALB, Auto Scaling, ECR), Docker, Kubernetes, Jenkins, CI/CD, OAuth2, JWT, Redis, ELK Stack, OpenTelemetry, Swagger, OpenAPI, JUnit, Mockito, Git, Agile Scrum, HIPAA Compliance.

Amway Corp ADA, MI **Sr. Java Developer | Jan 2021 - Sep 2022**
Roles & Responsibilities:

- Developed Spring Boot-based microservices for e-commerce and distributor management systems, supporting order processing, inventory management, and distributor operations with focus on scalability and maintainability.
- Contributed to refactoring legacy monolithic components into modular microservices, reducing code duplication and improving system modularity and maintainability.
- Built RESTful APIs using Spring Boot, implementing validation, exception handling, API versioning, and secure communication using OAuth2/JWT-based authentication.
- Implemented Kafka-based event-driven architecture for asynchronous communication between services, enabling order events, inventory updates, and notification workflows across distributed systems.
- Developed event-driven data processing workflows using Kafka for real-time order tracking and inventory synchronization across multiple backend services.
- Optimized complex PostgreSQL queries using joins, subqueries, and indexing strategies, improving performance of high-volume reporting and transactional operations.
- Worked with Hibernate ORM and Spring Data JPA, implementing entity relationships, custom repository methods, and pagination to improve database interaction efficiency.
- Implemented distributed tracing using Zipkin to monitor request flows across microservices, enabling identification of latency bottlenecks and improving system observability.
- Configured application monitoring using Prometheus for metrics collection and Grafana for visualization of API response times, error rates, and system health metrics.
- Integrated centralized logging using Splunk for log aggregation and analysis, enabling faster troubleshooting and root cause analysis of production issues.
- Participated in peer code reviews, ensuring adherence to coding standards, identifying defects, and improving overall code quality and maintainability.
- Contributed to CI/CD pipeline implementation using Jenkins, supporting automated builds, testing, and Docker-based deployments across environments.

- Containerized microservices using Docker, creating Dockerfiles and ensuring consistent deployment across development, testing, and production environments.
- Developed unit and integration tests using JUnit and Mockito, improving test coverage and reducing production defects through comprehensive testing practices.
- Created and maintained API documentation using Swagger/OpenAPI, documenting endpoints, request/response formats, and authentication requirements for team and external integration.
- Worked with senior architects and engineering teams to implement microservices design patterns including circuit breakers and service discovery for improved system resilience.

Environment: Java 8, Spring Boot, Microservices Architecture, RESTful APIs, Kafka, Event-Driven Architecture, Hibernate ORM, Spring Data JPA, PostgreSQL, SQL, JUnit, Mockito, Integration Testing, Unit Testing, Jenkins, CI/CD, Docker, Git, Agile Scrum, OpenAPI, Swagger, Prometheus, Grafana, Splunk, Zipkin, Code Review Practices, Performance Tuning.

Ford Motors, Dearborn, MI **Java Developer | Feb 2019 – Dec 2020**
Roles & Responsibilities:

- Developed backend services for automotive enterprise applications using Java 8 and Spring Boot framework, implementing business logic for distributed workflows supporting manufacturing and dealer operations systems.
- Contributed to microservices development as part of team initiative to decompose legacy monolithic systems into independently deployable services, improving system modularity and reducing deployment dependencies.
- Built RESTful APIs using Spring Boot with exception handling, request/response validation, and JSON payload support for integration between internal services and third-party automotive systems.
- Worked on implementing authentication and authorization using Spring Security framework with role-based access control (RBAC) and JWT token-based authentication to secure API endpoints and protect automotive data.
- Wrote optimized SQL queries with joins and subqueries in PostgreSQL, implemented database indexing for assigned modules, and supported performance tuning efforts to improve data retrieval times.
- Utilized Java 8 features including Streams API, Lambda expressions, Optional, and Method References to improve code readability and reduce boilerplate code in service implementations.
- Developed persistence layer components using Spring Data JPA with entity mappings, custom repository methods, and pagination support for efficient database operations.
- Worked with RabbitMQ for asynchronous message processing, implementing message consumers for background job workflows such as report generation and data synchronization tasks, improving system responsiveness by offloading long-running operations from synchronous API calls.
- Worked on containerizing microservices using Docker, creating Dockerfiles and supporting container configurations for deployment across development, staging, and production environments.
- Contributed to API documentation using Swagger/OpenAPI specifications, documenting request/response schemas and authentication requirements for frontend and third-party integration.
- Integrated backend services with Angular-based frontend applications, working with UI teams to implement API contracts and ensure proper data flow across application components.
- Supported CI/CD pipeline configuration using Jenkins, contributing to automated build processes, unit test execution, and Docker image creation for deployment automation.
- Developed unit and integration tests using JUnit and Mockito frameworks, contributing to improved test coverage and defect reduction across assigned modules.
- Performed debugging and troubleshooting of production issues, analyzed application logs, and worked with team members to implement fixes for system stability improvements.
- Managed source code using Git with GitFlow branching strategies, participated in code reviews, and collaborated on pull requests to maintain code quality standards.
- Collaborated with QA engineers, DevOps teams, and business analysts for requirement clarification, testing coordination, and deployment support.

Environment: Java 8, Spring Boot 2.x, Spring MVC, Spring Security, Spring Data JPA, Microservices Architecture, RabbitMQ, RESTful Web Services, JWT, OAuth2, Hibernate ORM, PostgreSQL, SQL, Java Streams API, Lambda Expressions, Method References, Optional, Angular 8+, Docker, Docker Compose, Dockerfile, Swagger, OpenAPI, Jenkins, CI/CD Pipelines, Maven, Git, GitFlow, JUnit 4, Mockito, Agile Scrum, JIRA.

OJAS Innovative Technologies, Hyderabad, India **Junior Software Developer | May 2014 – June 2017**
Roles & Responsibilities:

- Developed enterprise web applications using Java 8, Spring MVC, Hibernate, JSP, and Servlets for employee management, inventory tracking, and reporting systems.
- Assisted in implementing Spring MVC architecture with Controller-Service-DAO layers and Dependency Injection using XML and annotation-based configuration.

- Developed and contributed to RESTful web services using JAX-RS (Jersey) for internal application integration and JSON/XML-based data exchange.
- Built user interfaces using HTML5, CSS3, JavaScript, jQuery, and AngularJS (1.x) with form validations, AJAX calls, and dynamic UI interactions.
- Created and optimized SQL queries with joins and contributed to developing PL/SQL stored procedures in Oracle 11g for data processing and reporting.
- Worked with Hibernate ORM mappings and assisted in implementing entity relationships and lazy loading for database interactions.
- Developed unit test cases using JUnit and contributed to integration testing using Mockito to support code quality.
- Contributed to integrating third-party libraries such as Apache POI (Excel), JasperReports (PDF), and JavaMail (email notifications).
- Used Maven for build management and assisted in resolving dependency issues and supporting application builds.
- Supported deployment on Apache Tomcat servers, assisting in configuration and release activities across development and QA environments.
- Assisted in implementing application logging using Log4j and supported troubleshooting through log analysis.
- Contributed to basic database performance improvements by optimizing queries and supporting indexing efforts.
- Used Git for version control with feature branching and participated in code reviews through pull requests.
- Participated in Agile Scrum ceremonies including sprint planning, daily stand-ups, backlog grooming, and retrospectives.

Environment: Java 8, Spring MVC, Hibernate, JSP, Servlets, JAX-RS (Jersey), RESTful APIs, AngularJS (1.x), HTML5, CSS3, JavaScript, jQuery, Oracle 11g, SQL, PL/SQL, Apache Tomcat, Maven, Git, JUnit, Mockito, Apache POI, JasperReports, JavaMail, Log4j, Agile Scrum.

EDUCATION:

Master's in computer science — Aug 2017 - Dec 2018
Saint Louis University — 3.7 GPA